

torch.as_tensor#

https://pytorch.org/docs/stable/generated/torch.as_tensor.html#torch.as_tensor

Description:

Converts data into a tensor, sharing data and preserving autograd history if possible. If data is already a tensor with the requested dtype and device then data itself is returned, but if data is a tensor with a different dtype or device then it's copied as if using `data.to(dtype=dtype, device=device)`. If data is a NumPy array (an ndarray) with the same dtype and device then a tensor is constructed using `torch.from_numpy()`. If data is a CuPy array, the returned tensor will be located on the same device as the CuPy array unless specifically overwritten by device or a default device. The device of the CuPy array is inferred from the pointer of the array using `cudaPointerGetAttributes` unless device is provided with an explicit device index.

Function Signature

```
torch.as_tensor(data: Any, dtype: Optional[dtype] = None,  
device: Optional[DeviceLikeType]) → Tensor#
```

Parameters

Code Examples

```
>>> a = numpy.array([1, 2, 3])  
>>> t = torch.as_tensor(a)  
>>> t  
tensor([ 1,  2,  3])  
>>> t[0] = -1  
>>> a  
array([-1,  2,  3])  
  
>>> a = numpy.array([1, 2, 3])  
>>> t = torch.as_tensor(a, device=torch.device('cuda'))  
>>> t  
tensor([ 1,  2,  3])  
>>> t[0] = -1  
>>> a  
array([1,  2,  3])
```

Notes & Warnings

NOTE

See also `torch.tensor()` never shares its data and creates a new “leaf tensor” (see Autograd mechanics).

Detailed Documentation

Documentation

Converts data into a tensor, sharing data and preserving autograd history if possible.

If data is already a tensor with the requested dtype and device then data itself is returned, but if data is a tensor with a different dtype or device then it's copied as if using `data.to(dtype=dtype, device=device)`.

If data is a NumPy array (an ndarray) with the same dtype and device then a tensor is constructed using `torch.from_numpy()`.

If data is a CuPy array, the returned tensor will be located on the same device as the CuPy array unless specifically overwritten by device or a default device. The device of the CuPy array is inferred from the pointer of the array using `cudaPointerGetAttributes` unless device is provided with an explicit device index.

`torch.tensor()` never shares its data and creates a new “leaf tensor” (see Autograd mechanics).

data (array_like) – Initial data for the tensor. Can be a list, tuple, NumPy ndarray, scalar, and other types.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, infers data type from data.

device (torch.device, optional) – the device of the constructed tensor. If None and data is a tensor then the device of data is used. If None and data is not a tensor then the result tensor is constructed on the current device.

